09/08/2024

Study of different types of Network cables

Aim : To study the diff. types of ~~cables~~ network cables used by us,

(a) Understand different types of network cable

The diff. types of cables used in networking are :-

(i) Unshielded Twisted Pair (UTP) cable

(ii) Shielded Twisted Pair (STP) cable

(iii) Coaxial Cable

(iv) Fibre Optic Cable

Cable Type	Category	Maximum Data Transmission	Advantages / Disadvantages	Application / Use
UTP	Category 3	10 bps	Advantages * Cheaper in cost. * Easy to install as they have a smaller overall diameter. Disadvantages * More prone to EMI & noise	10 Base-T Ethernet
	Category 5	Upto 100 Mbps		Fast Ethernet, Gigabit Ethernet
	Category 5e	1 Gbps		

STP	Category 6, 6a	10 Gbps	Advantages * Shielded * Faster than UTP. * less susceptible to noise & interference. Disadvantages * Expensive * Greater installation effort.	Gigabit Ethernet, 10G Ethernet (55 m) Widely used in data centres
SS TP	Category 7.	10 Gbps		Gigabit Ethernet, 10G Ethernet (100 m)
Coaxial Cable	RG-6 RG-59 RG-11	10-100 Mbps	Advantages * High bandwidth * Immune to interference * low loss bandwidth * Versatile Disadvantages * Limited distance * Cost * Size is bulky	Speed of signal is 500m television network * High speed internet connections

	Fibre Optics Cable	Single mode, Multi mode	100 Gbps	Advantages *High speed *High bandwidth *High security *Long distance Disadvantages * Expensive * Requires skilled installers	Max. distance of fibre optics cable is around 100 meters.

(b) Make Your Own Ethernet Cross-Over Cable / Straight Cable :-

Tools and parts needed :-

* Ethernet Cabling : CAT5e is certified for gigabit support, but CAT5 cabling works as well, just over shorter distances.
* A crimping tool : This is an all-in-one networking tool shaped to push down the pins in the plug & strip and cut the shielding off the cables.
* Two RJ45 plugs.
* Optional two plug shields

Result : Hence, the diff. types of network cables used by us are tested and connected together.

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